Worksheet: Multivariable Optimization

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Consider the function

$$f(x, y) = x^2 + xy + y^2 - 3x$$

(a) Find f_x and f_y, and use the first-order conditions to find the critical point.

(b) Find f_{xx}, f_{yy}, and f_{xy}.

(c) Is the critical point a maximum, a minimum, or neither? What is the value of f there?

A firm sells two products in quantities Q_1 and Q_2, and its profit is

$$\pi(Q_1, Q_2) = 12Q_1 + 18Q_2 - 2Q_1^2 - Q_1Q_2 - 2Q_2^2$$

(a) Write the first-order conditions and solve for the output levels Q_1^* and Q_2^*.

(b) Use the second-order conditions to check that this is a maximum. What is the maximum profit?